AAYUSH KHANAL
Kathmandu, Nepal | aayushkhanal.official@gmail.com | https://www.linkedin.com/in/khanalaayush | +977-9865187774 |

Career Objective

An aerospace engineering student passionate about aerodynamics, research, design, and propulsion offering analytical skills to solve problems. Eager to gain hands-on experience with professionals and grow professionally within the aerospace industry.

Key Skills

- **Technical Skills**: Computational Fluid Dynamics Simulation, Aerodynamic Design, System Design & Prototyping
- **CFD Tools**: ANSYS FLUENT, SU2, OPENFOAM
- **Design Tools**: CATIA V5, SOLIDWORKS, Pointwise, XFLR5, Plane Maker (X-Plane 12), OpenVSP
- **Programming Tools**: MATLAB, Arduino
- **Documentation & Visualization Tools**: LaTeX, MS Products, Inkscape
- **Soft Skills**: Problem Solving, Team Collaboration, Project Management, Adaptability

Education

Pulchowk Campus — **2021 – 2025**
Bachelor of Aerospace Engineering, Cumulative Percentage: 79.84 % — *Lalitpur, Nepal*

Aroma College of Applied Science And Management — **2017 – 2019**
Higher Secondary Education (Grades 11 and 12), CGPA: 3.65 — *Chitwan, Nepal*

Shiksha Bikash Higher Secondary School — **2016 – 2017**
Secondary Education Examination , GPA: 3.75 — *Chitwan, Nepal*

Experience

Indian Institute of Technology, Kharagpur — **October 2024 – Decemeber 2024**
Research Intern | CATIA V5, MATLAB, Pointwise, SU2, ANSYS FLUENT — *Kharagpur, West Bangal*

- Investigated the effect of drag reduction with the addition of conventional aerospike in bluff body at SU2 and ANSYS FLUENT solvers.
- Experimentally verified the shock angles obtained at Mach 2.43 using schlieren images.

Projects

Addition of Heat Flux in a Closed Cavity | *OPENFOAM* — **March 2025**

- Investigated the effect of varying wall lengths on transient natural convection utilizing *buoyantPimpleFoam*.
- Found out the increment in heater wall length increased the pressure, velocity, temperature, and heat transfer inside the body.

Development of Subsonic Wind Tunnel with Gust Generator | *CATIA V5, MATLAB, Pointwise, ANSYS FLUENT* — **May 2024 – January 2025**

- Designed and developed a subsonic wind tunnel with a contraction ratio of 6:1 and integrated a gust generator using oscillating vanes.
- Achieved 0.88 % turbulent intensity with a mean velocity of 8.6 m/s at the test section, and 2.26 % turbulent intensity at the outlet.
- Developed a smoke visualization setup to observe produced longitudinal gusts.
- Investigated the effect of design parameters (chord length, spacing, frequency, and degree of oscillation) on gust generator's performance.

Proposed Modification of B-52H named ATOM for Air-Launch to Orbit | *XFLR5, X-Plane 12, MATLAB* — **June 2024 – September 2024**

- Optimized for high altitude performance (35,000 – 45,000 feets) with 10 Aspect Ratio and 6 Rolls-Royce Trent XWB-84 for 2.141 MN thrust.
- Designed for cruising speed of Mach 0.86 for 6000 km range having 12,700 kg payload weight and 92,505 kg as maximum take off weight.
- Marked a Cooper-Harper rating of 4, with stable damping of 150 seconds for dutch roll, 500 seconds for phugoid, 10 seconds for short-period, and 50 seconds for spiral mode tests.
- Estimated the cost of aircraft manufacturing as $141 million with $9.11 million dollars per launch, including fuel, maintenance, crew, and logistics.

Investigation of Drag on Geometric Bodies (Rectangle, Bullet, and Airfoil) | *CATIA V5, ANSYS FLUENT* — **June 2024**

- Compared the drag coefficients of the bodies to understand their performance under same condition.
- Found that the rectangle had the highest drag coefficient, airfoil less c_d, and bullet body was in the middle range.

Investigation of airfoil thickness at low Reynolds Number | *CATIA v5, ANSYS FLUNET* — **October 2023**

- Performed CFD simulations on NACA 0012, 4412, and 4415 with varying thicknesses (50%, 100%, and 125%) utilizing ANSYS FLUENT.
- Found that the increment in thickness increased lift with drag, and 100 % thickness offered best balance for optimizing aerodynamic performance.

Design and Development of a Glider | *XFLR5, CNC Foam Cutter* — **January 2022**

- Designed a glider with analysis of airfoil Eppler 205 in XFLR5 with 8 Aspect Ratio and 3 meter wingspan ensuring optimal center of gravity placement.

Miscellaneous

Designed Lockheed U-2 in Plane Maker and performed dynamic stability tests | *XFLR5, MATLAB, X-Plane 12* — **January 2024 – March 2024**

Scaled F-15 Model for manufacturing | *CATIA V5* — **January 2024 – February 2024**

Certificates

ANSYS Workshop | *Link*
Aerodynamic Shape Optimization | *Link*
NEPAL ENGINEERING COUNCIL | *95265*
IELTS TEST | *O: 7.5, R: 8, L: 8, W: 6.5, S:7.5*